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AI Engineer & Edge AI Engineer

📍 Ho Chi Minh City | 📞 0935 019 948 | ✉️ dat614943@gmail.com | 🌐 [in/dat614943](https://in.dat614943) | 🏠 github.com/dat2003as

PROFESSIONAL SUMMARY

AI Team Lead and **Edge AI Engineer** specializing in architecting commercial-grade computer vision pipelines and migrating restrictive architectures to license-free solutions (RTMDet, PPYOLOE-R). Proven expertise in deploying server-side vision modules for real-world constraints, while actively driving **R&D initiatives** to compile highly optimized execution graphs (.engine, .cix) for edge hardware (**NVIDIA Jetson Nano, Orange Pi 6 Plus**).

TECHNICAL SKILLS

Hardware & Edge AI	NVIDIA Jetson Nano, Orange Pi 6 Plus (CIX 45 TOPS), TensorRT (.engine) , ONNX → .cix , Hardware-aware Optimization.
Deep Learning Stack	PyTorch, PPYOLOE-R , RTMDet , YOLOX , EfficientNet, timm, Metric Learning, ONNX Runtime, CBAM Attention.
Languages & Tools	GStreamer , OpenCV , Python , Shell Scripting, Embedded Linux, Docker, ZeroMQ, FastAPI.
Generative AI	RAG Architectures, LangChain, Qdrant (Vector DB), Local LLM Integration (Llama 3.2).

PROFESSIONAL EXPERIENCE

AI Team Lead

Mebisoft

Nov 2025 – Present

Ho Chi Minh City

- **Spearheaded a server-side aquaculture vision pipeline**, strategizing the migration from restrictive YOLO models to Apache 2.0/license-free architectures (RTMDet, PPYOLOE-R, YOLOX) for commercial deployment.
- **Engineered a high-density object counting module** (1024x1024 input) by orchestrating cross-functional mobile UI constraints for fixed-aspect ratio cropping, boosting real-world accuracy to **90%** for sparse and **>80%** for highly overlapped environments.
- **Formulated a production-ready measurement system** fusing RTMDet-Seg with ArUco-based skeletal geometry, achieving a strict **<10% measurement error** against physical benchmarks in real-world operations.
- **Engineered a hardware-aware compilation pipeline** for high-density aquaculture counting, translating PPYOLOE-medium into optimized **TensorRT (.engine)** graphs. Validated architecture scalability by securing a **>120 FPS** baseline on RTX 3050Ti to derisk throughput constraints prior to physical Jetson Nano deployment.
- **Developed a Proof-of-Concept (PoC) anomaly detection pipeline** using sequential segmentation and EfficientNet-B0 with CBAM attention, securing **94% accuracy on test sets** (production deployment pending long-term farm data collection).
- **Spearheaded the deployment of an enterprise RAG chatbot (Mebieco)**, leveraging Vanna and pgvector for advanced Text-to-SQL capabilities; implemented JWT-based security and SQL guardrails, enabling reliable intent-driven querying across **60+ PostgreSQL tables** for aquaculture monitoring.

Edge AI Engineer

Be-Earning

May 2025 – Oct 2025

[Repo](#) | Ho Chi Minh City

- **Accelerated inference throughput by 3×** vs. CPU baseline by engineering a custom ONNX-to-.cix model compilation pipeline targeting the 45 TOPS CIX CD8180 NPU.
- **Engineered a low-latency video ingestion pipeline** via GStreamer and adaptive frame skipping, decoupling capture from AI inference. Reduced input lag to **<50ms**, ensuring smooth tracking at 5–7 FPS on embedded constraints.
- **Optimized multi-task biometric models** (PyTorch Lightning) using MixUp and class-weighting to mitigate data imbalance, securing **>85% validation accuracy** under strict edge memory constraints.
- **Implemented Adaptive Recognition Logic** fusing CCCD, Face ID, and Body Re-ID vectors via multi-index FAISS retrieval, achieving **<1.5s latency** for billion-scale database matches.

NOTABLE PROJECTS

Enterprise RAG Microservices System

[Repo](#) | Runner-up, NTTU AI Comp. 2025

- Designed a scalable RAG microservices architecture handling concurrent requests via Kafka and Docker Compose; integrated Llama 3.2 and Qdrant Vector DB.

Ultra-Low Latency Gesture Control

[Repo](#) | Innovation Award, NTTU AI Comp. 2024

- Built a real-time HCI system (**<50ms latency**) by optimizing serial communication between a Python CV pipeline and Arduino micro-controllers.

EDUCATION

Bachelor of Information Technology

Nguyen Tat Thanh University

2021 – 2025

GPA: 3.47/4.0